MG MUSALLAM GROUP · DELIVERY OPERATIONS

# MGHub

## System Architecture & Technical Overview

The platform end to end — business modules, the Firestore data model, the payroll and synchronisation engines, every callable, route, cron and scheduled job, and the feature matrix that maps them to the screens operators use.

Next.js 16 · Firestore | Cloud Functions gen2 · me-central1 | Cloudflare DO · R2

Read from the MG-GO-Admin and MG-GO source trees · No placeholders · English edition

Version 1.0 · 13 Nov 2026

MGHUB · SYSTEM ARCHITECTURE & TECHNICAL OVERVIEW

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FIGURES AND PLATES


1 The platform in four layers
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3 Session to permission in one pass
4 Two engines, deliberately separate: the month grid decides what a day was,


| VERSION | READER | REBUILD |
|---|---|---|
| Version 1.0 · 13 Nov 2026 | Client technical stakeholders, architects, onboarding engineers | npm run docs:architecture |

**Platform:** MGHub Admin & Operations Portal (Musallam Group / DPD) **Document version:** 1.0 · **Generated:** 13 November 2026 **Sources:** `src/app`, `src/features`, `src/lib`, `functions/src`, `infra/workers/dpd-live`, `supabase/migrations`, `vercel.json` **Method:** every table, route, function and cron below was read from the repository. Nothing is illustrative.

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# 1 Executive Summary & Core Modules

## 1.1 What the platform is

MGHub is the back-office control tower for a contracted rider workforce: roughly 950 delivery riders across Kuwait, dispatched by third-party partners (Americana, Keeta, Talabat, Uber Eats, Door Dash, Five Guys). **Order allocation is external** — the partner apps decide which rider gets which order. MGHub owns everything around that:

- who the rider is (identity, documents, contracts, residency)
- whether they were present and productive (shift, attendance, GPS, zone discipline)
- what they are paid (per-order incentives, monthly payroll, HR payroll)
- what the client can see (a scoped partner portal)
- the operational surfaces built on top (e-signature, visit bookings, accident cases, fleet maintenance, notifications)

Two clients consume it:

| Client | Repository | Surface |
|---|---|---|
| Admin / staff | `MGgo-Admin` (this repo) | Full back-office web app |
| Rider | `MG-GO` (separate repo) | Flutter app, OTP + passcode login |

A third surface — the Cloudflare Worker `dpd-live` — sits between the rider phones and the admin live map.

## 1.2 Business modules (route inventory, from `src/app/[locale]/(dashboard)`)

| # | Module | Route | Permission | Purpose |
|---|---|---|---|---|
| 1 | **Dashboard / Launcher** | `/dashboard` | `dashboard.view` | Dark launcher grid; the hub of the app |
| 1b | Control Tower | `/dashboard/ops` | `dashboard.view` | KPI snapshot across modules |
| 2 | **Live Tracking V1** | `/live-tracking` | `live_tracking.view` | Durable pin per rider from `driver_locations` |
| 2b | **Live Tracking V2** | `/live-tracking-v2` | `live_tracking.view` | 1 Hz deck.gl canvas over a Cloudflare Durable Object |
| 3 | **Employees** | `/drivers`, `/drivers/[id]`, `/drivers/new`, `/drivers/groups`, `/drivers/dashboard` | `drivers.view` / `drivers.manage` | Rider roster, intake → approval, documents, device link |

| # | Module | Route | Permission | Purpose |
|---|---|---|---|---|
| 4 | **Fleet Hub** | `/fleet` → `/vehicles`, `/fuel`, `/assets`, `/fleet/service`, `/fleet/handover`, `/fleet/accident`, `/fleet/rj` | `vehicles.*`, `fuel.*`, `assets.*`, `service.*` | Vehicles, fuel ledger, assets, workshop, handovers, accidents |
| 5 | **Deliveries** | `/deliveries`, `/deliveries/reconciliation`, `/deliveries/zones` | `deliveries.view` / `deliveries.manage`, `order_recon.*` | Verify / reject orders, AM-vs-MGGO comparison |
| 6 | **Payroll Hub** | `/payroll-hub` → `/payroll`, `/payroll/combined`, `/payroll/attendance-orders`, `/payroll/hr`, `/payroll/settings`, `/payroll/pay-plan` | `payroll.*` (10 slugs) | Monthly payroll, HR sheet, rules engine |
| 7 | **Operations Hub** | `/operations` → `/operations/roster` (+ `/roster/upload`, `/roster/changes`, `/roster/alerts`, `/roster/report`) | `attendance.*`, `earnings.*`, `drivers.manage` | Rule tables, shift roster, attendance alerting |
| 8 | **Attendance** | `/attendance` | `attendance.view` / `attendance.manage` | Live presence, logs, exceptions, analytics |
| 9 | **Notifications** | `/notifications` (+ `/new`, `/[id]`, `/history`, `/automations`, `/templates`, `/analytics`) | `notifications.*` (5 slugs) | Campaigns, audiences, FCM dispatch, automations |
| 10 | **Roles & Permissions** | `/settings/roles`, `/settings/staff-access` | `roles.manage`, super admin | 4-level access editor + role matrix |
| 11 | **EmployeeDesk** | `/employeedesk` (+ `/esign/*`, `/visits/*`, `/all`, `/[id]`, `/reports`, `/settings`) | `requests.*`, `esign.*`, `visits.*` | Request & Complaint (RCM), e-signature, visit bookings |
| 12 | **Visit Slots** | `/visit-bookings` (+ `/all`, `/calendar`, `/reception`, `/slots`, `/departments`, `/branches`, `/reports`) | `visits.*` (4 slugs) | Appointment booking for HR/compliance desks |
| 13 | **Performance** | `/performance`, `/performance/settings` | `performance.*` (5 slugs) | SOP v2.0 OPD/DPD/UTR analytics |

| # | Module | Route | Permission | Purpose |
|---|---|---|---|---|
| 14 | **Client Portal** | `/client-portal-v2` (+ `/employees`, `/clients`, `/new`, `/preview/[projectKey]`) | `client_portal.view` | Partner-scoped read-only dashboard |
| 15 | **System Settings** | `/settings` (+ 24 sub-pages) | `settings.*` | Branding, menus, languages, catalogs, storage, cleanup |
| 16 | Support / Assistant / Compliance / Wrong actions / Document expiry / Driver devices / Payouts / Earnings / DPD rules | `/support`, `/assistant`, `/compliance`, `/wrong-actions`, `/document-expiry`, `/driver-devices`, `/payouts`, `/earnings`, `/delivery-rules`, `/incentive-rules` | per module | See §5 |

41 top-level route folders under `src/app/[locale]/(dashboard)`, **92 permission slugs** in `src/lib/auth/permission-catalog.ts`.

## 1.3 Tech stack

| Layer | Choice | Where |
|---|---|---|
| **Framework** | Next.js **16.2.6**, App Router, Turbopack, React Server Components | `package.json`, `next.config.ts` |
| **Language** | TypeScript (strict), no `.js` app sources | `tsconfig.json` |
| **Styling** | Tailwind CSS v4 + shadcn/ui primitives on CSS custom properties; single locked navy/green theme | `src/app/globals.css`, `src/app/themes.css`, `src/components/ui/*` |
| **Product components** | `src/components/app/*` (`AppPage`, `AppListCard`, `AppDataTable`, `AppModalFooter`, `TableToolbar`, `AppHubShell`) | shared shell for every page |
| **Server state** | **TanStack Query v5** — one `QueryClient` per browser tab; `staleTime` 60 s, `gcTime` 5 min, `refetchOnWindowFocus: false` | `src/lib/query/*`, `src/components/providers/query-provider.tsx` |
| **Client state** | React hooks + Riverpod-equivalent pattern via context providers; no Redux | `src/contexts/*` |
| **i18n** | **next-intl**, locales `en` + `ar` (RTL), **12,625 keys × 2** with a parity guard | `src/messages/*.json`, `scripts/check-messages.mjs` |
| **Database** | **Firestore** (named database `default`), Admin SDK server-side only | `src/lib/firebase/*`, `functions/src/core/fs.ts` |
| **Auth** | Firebase Auth → **session cookie** (`__session`) verified in `src/proxy.ts`; custom claims carry `superAdmin` | `src/lib/firebase/session.ts`, `src/proxy.ts` |
| **Backend** | **Cloud Functions gen2** (`firebase-functions/v2`) in `me-central1` / `musallam-delivery-prod` | `functions/src` |

| Layer | Choice | Where |
|---|---|---|
| **Real-time (maps)** | **Cloudflare Durable Object** `FleetRoom` + WebSocket, with a Supabase-Broadcast-era poll fallback | `infra/workers/dpd-live` |
| **Object storage** | **Cloudflare R2** `dpd-private-prod` (driver docs, e-sign PDFs, fuel, request attachments, GPS archive) | `src/lib/storage/r2-client.ts` |
| **Hosting** | **Vercel**, region `bom1`, `maxDuration` 300 s on the dashboard layout, the order-recon routes and `/deliveries/reconciliation` | `vercel.json`, `(dashboard)/layout.tsx` |
| **Charts** | **Recharts** (Performance trend only) + hand-built CSS/SVG bars elsewhere | `src/features/performance` |
| **Maps** | Google Maps JS + `@vis.gl/react-google-maps`; **deck.gl** for V2; Leaflet for `/zones` editor | `src/features/driver-tracking`, `src/features/live-tracking-v2`, `src/features/zones` |
| **Excel / CSV** | **ExcelJS** (styled workbooks) + **xlsx** (namespace import) + custom CSV writers | `src/lib/*xlsx*`, `src/lib/csv/*` |
| **Push** | **Firebase Cloud Messaging** (FCM) via Admin SDK | `functions/src/rpcs/driver-push-tokens.ts`, `notifications.ts` |
| **PDF** | Chrome print for e-sign sheets; `pdf-lib` for signature stamping | `functions/src/rpcs/esign-compose*` |
| **Observability** | Sentry (client + functions), `perf` JSON logs from the callable router | `src/lib/sentry/*`, `functions/src/perf/*` |

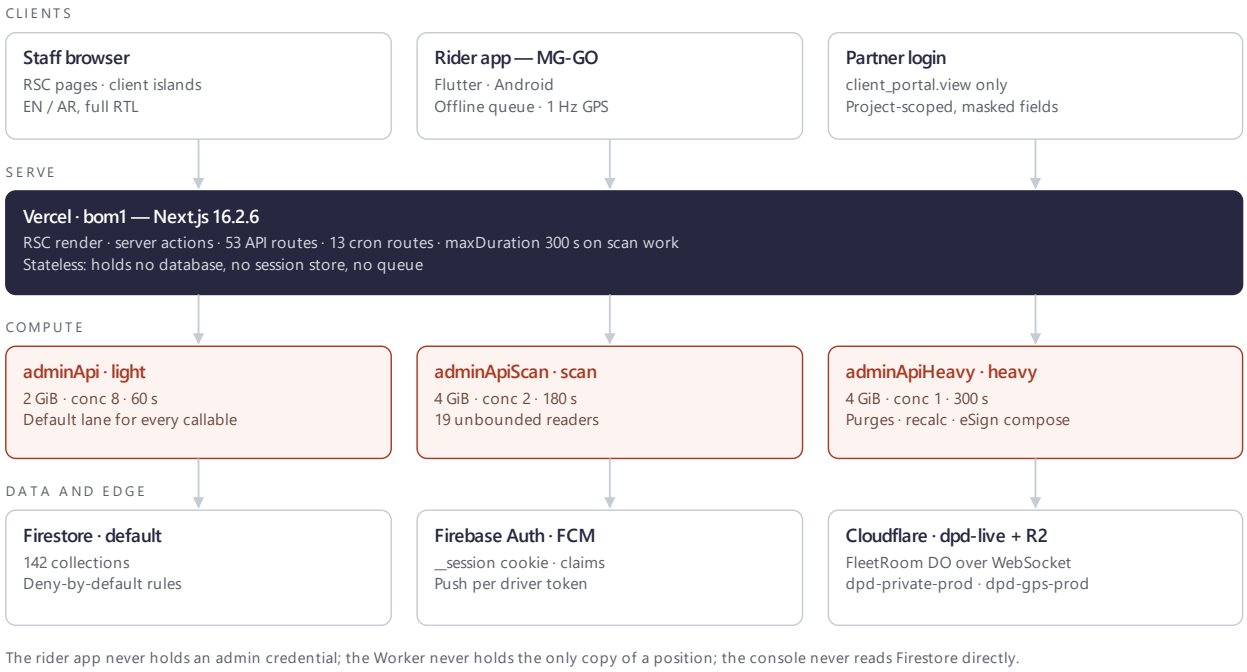



**Figure 1.** The platform in four layers. Every arrow is a real network hop: an HTTPS callable, a WebSocket frame, or an Admin SDK write.

# 2 Database Schema & Data Models

## 2.1 Two schemas, one live

The project migrated from **Supabase Postgres** to **Firestore**. Both exist in the tree; only one is live.

| Schema | Status | Location | Count |
|---|---|---|---|
| **Firestore collections** | **Live** — every read and write goes here | `src/lib/firebase/db.ts`, `functions/src/core/collections.ts` | **142 named collections** (+ 6 `uniq_*` lock collections, `counters`) |
| Postgres migrations | **Historical** — the Supabase project `eoksxkdssptgyqyywdju` is paused; admin and Worker no longer call it | `supabase/migrations/*.sql` | 398 files |

> **Read this before writing a query.** A Postgres table name in a migration does **not** mean a live table. `deliveries`, `drivers`, `requests`, `esign_requests` and the rest are Firestore collections now. `src/types/database.ts` is a generated artefact kept for type continuity, not a live schema.

## 2.2 Identity, access and configuration

| Collection | Purpose | Key fields | Relationships |
|---|---|---|---|
| `profiles` | Auth mirror + panel access | `id` (= `auth.uid`), `role` (`staff` \| `rider`), `approval_status` (`pending` \| `approved` \| `rejected`), `admin_role_id`, `access_kind` (`manager` \| `user` \| `partner`), `partner_project_key`, `staff_department`, `sessions_revoked_at`, `archived_at` | 1:1 `drivers`; N:1 `admin_roles` |
| `drivers` | The rider record | `employee_id` (unique, 4–8 digits), `driver_code` (5 digits), `client_id`, `phone`, `civil_id`, `passport_no`, `zone_id`, `partner_id`, `vehicle_id`, `project_key`, `status`, `app_passcode`, `custom_fields` (jsonb), `current_odometer_km`, `avatar_object_key` | N:1 `zones`, `partners`, `vehicles`; M:N `restaurants` via `driver_restaurants` |
| `driver_intakes` | Add-Driver draft before approval | same identity fields + `linked_profile_id`, `archived_at` | 1:1 `drivers` after approve |
| `admin_roles` | Panel role | `slug`, `name`, `is_super_admin` | 1:N `admin_role_permissions` |
| `admin_permissions` | Slug catalog (172 slugs) + `catalog` doc | `slugs: string[]` | — |

| Collection | Purpose | Key fields | Relationships |
|---|---|---|---|
| `admin_role_permissions` | Role → slugs | `permission_slugs: string[]` | N:1 `admin_roles` |
| `admin_user_permissions` | Per-user ticks | `permission_slugs: string[]` | 1:1 `profiles` |
| `admin_ui_preferences` | Per-user column order / view | `user_id`, `preference_key`, `value` ( `order`, `visible`, `sort`, `groupBy`, `viewMode` ) | N:1 `profiles` |
| `admin_role_ui_defaults` | Per-role column defaults | `role_id`, `preference_key`, `value` | N:1 `admin_roles` |
| `admin_activity_logs` | Audit trail | `action`, `entity_type`, `entity_id`, `before`, `after`, `actor_id`, `page_path` | N:1 `profiles` |
| `auth_epochs` | Cache-bust counters ( `current` ) | `access`, `settings`, `menu`, `payroll`, … | — |
| `menu_configs` | Per-role sidebar tree | `admin_roles.slug` → JSON tree | N:1 `admin_roles` |
| `locales` | Enabled locales | `code`, `is_enabled`, `is_default`, `dir` | — |
| `app_settings` | Singleton doc `1` | branding ( `app_name`, `logo_url`, `font` ), `maintenance_mode`, `super_admin_claimed`, `super_admin_user_id`, force-update block, `pickup_auto_cancel_hours`, `attendance_auto_checkout_minutes`, `delivery_ontime_minutes`, `performance_score_weights`, `incentive_band_math_from` | — |
| `counters` | 8 code sequences | `driver_code_seq`, `restaurant_code_seq`, `request_code_seq`, `visit_booking_code_seq`, `esign_code_seq`, `appointment_code_seq`, `fuel_refund_code_seq`, `esign_batch_code_seq` | — |
| `uniq_employee_id` · `uniq_phone` · `uniq_civil_id` · `uniq_passport` · `uniq_plate` · `uniq_passcode` | Lock documents standing in for partial unique indexes | one doc per held value | — |

**Constraints that live in code, not in a schema** (Firestore has no DDL):

| Rule | Enforced at |
|---|---|
| `employee_id` unique among live riders, 4–8 digits | `driver-uniqueness.ts`, `uniq_employee_id`, `admin_approve_driver` |

| Rule | Enforced at |
|---|---|
| `phone` / `civil_id` / `passport` optional but unique when present | partial-index lock docs; `''` unrepresentable |
| `driver_code` exactly 5 digits from `driver_code_seq` | `driver-code.ts` |
| `app_passcode` 6 digits, minted when `status` becomes `active` | `admin_approve_driver` |
| Kuwait day stored as `YYYY-MM-DD` string, never derived at read | `kuwaitDayString()` in `db.ts` |

## 2.3 Workforce, operations and delivery

| Collection | Purpose | Key fields | Relation ships |
|---|---|---|---|
| `partners` | Dispatch partners = **Clients** in the UI | `name`, `slug`, `logo_url`, `description` | 1:N `restaurants`, `drivers` |
| `restaurants` | Merchant store | `name`, `partner_id`, `zone_id`, `latitude`, `longitude`, `status`, `external_merchant_id` | N:1 `partners`, `zones`; M:N `drivers` |
| `driver_restaurants` | Live rider ↔ store assignment (M:N) | `driver_id`, `restaurant_id` | junction |
| `driver_intake_restaurants` | Intake-stage assignment | same | junction |
| `restaurant_geofences` | Per-store inclusion / exclusion shapes | `restaurant_id`, `kind`, `zone_type`, `geometry` | N:1 `restaurants` |
| `zones` | Delivery areas | `name`, `code`, `zone_type`, `geometry` (GeoJSON polygon/circle), `color` | 1:N `restaurants`, `drivers` |
| `vehicles` | Fleet record | `reg_number` (plate = Vehicle ID, unique), `bike_id` (slug), `vehicle_type_key`, `vehicle_company_id`, `vehicle_use_type`, `condition`, `status`, `current_driver_id`, `current_odometer_km`, `chassis_no`, `model_year`, `company_id` | 1:1 `drivers` (current holder) |

| Collection | Purpose | Key fields | Relationships |
|---|---|---|---|
| `vehicle_types` · `vehicle_companies` · `vehicle_use_types` | Fleet catalogs | `key`, `label_en`, `label_ar`, `is_active` | — |
| `deliveries` | One order | `external_order_id`, `driver_id`, `restaurant_id`, `zone_id`, `partner_id`, `status` (`pending` \| `in_transit` \| `pending_verification` \| `verified` \| `rejected` \| `cancelled`), `created_at`, `pickup_at`, `started_to_restaurant_at`, `delivered_at`, `shift_date`, `proof_keys`, `rejection_reason` | N:1 `drivers`, `restaurants` |
| `rollups_driver_day` | Pre-aggregated per rider-day | `{driverId}_{YYYY-MM-DD}`, `pending`, `in_transit`, `cancelled`, `attendance_present`, `zone_id`, `partial` | derived |
| `rollups_zone_month` | Pre-aggregated per zone-month | `{zoneId}_{YYYY-MM}`, orders, rider days | derived |
| `attendance_logs` | One check-in | `driver_id`, `log_date`, `check_in_at`, `check_out_at`, `check_out_reason` (`manual` \| `auto_shift_end` \| `auto_timeout`), `status`, denormalised identity | N:1 `drivers` |
| `driver_sessions` | Duty session | `driver_id`, `is_online`, `went_online_at`, `ended_at`, `device_id` | N:1 `drivers` |
| `driver_daily_shifts` | Rider-submitted or roster-uploaded shift | `driver_id`, `shift_date`, `start_at`, `end_at`, `source` (`roster` \| `driver`), `approved_change_id` | N:1 `drivers` |
| `roster_changes` | Requested schedule change | `driver_id`, `shift_date`, `status`, `decided_by` | N:1 `driver_daily_shifts` |
| `driver_off_structure` | Monthly OFF entitlement | `{driverId}_{YYYY-MM}`, `off_days` (default 2) | N:1 `drivers` |
| `driver_leave_plans` | Leave ledger | `driver_id`, `leave_type`, `start`, `end`, `days`, `status`, `year_month`, `source` | N:1 `drivers` |

| Collection | Purpose | Key fields | Relation ships |
|---|---|---|---|
| `driver_locations` | Latest fix, one doc per rider | `driver_id` (PK), `lat`, `lng`, `speed_mps`, `heading_deg`, `accuracy_meters`, `zone_status`, `out_of_zone_since`, `last_seen_at`, `last_report_at`, `distance_today_meters`, `day` | 1:1 `drivers` |
| `driver_location_events` | 90-day position history | append-only | N:1 `drivers` |
| `driver_operation_events` | Server-authored audit of every rider RPC | `driver_id`, `event_type`, `payload`, `expire_at` (30 d) | N:1 `drivers` |
| `driver_telemetry_events` | Client-authored phone telemetry | `event_id` (idempotency), `event_type`, sanitised `context`, `expire_at` (14 d) | N:1 `drivers` |
| `rider_day_stats` | Per-day GPS totals | `{driverId}_{day}` | — |
| `fleet_events` | Edge-derived Class B events ( `overspeed`, `sustained_idle`, `low_battery`, `zone.exit`, `range.exit`, `gps.offline`, `shift.late`, `shift.overrun` ) | `driver_id`, `type`, `at`, `expire_at` (400 d) | N:1 `drivers` |
| `rider_signals` | Screen-refresh counters for the app | `{uid}` → `home_rev`, `earnings_rev`, … | 1:1 `drivers` |
| `driver_push_tokens` | FCM tokens | `driver_id`, `token`, `is_active`, `last_seen_at` | N:1 `drivers` |
| `driver_device_sessions` | Device binding + build reporting | `driver_id`, `device_id`, `app_version`, `device_meta`, `device_meta_at` | N:1 `drivers` |
| `driver_shifts` (via `driver_daily_shifts` ) · `driver_devices` (via `driver_device_sessions` ) | see above | | |

## 2.4 Money: earnings, payroll, recon

| Collection | Purpose | Key fields | Relationships |
|---|---|---|---|
| `delivery_rules` | Per-delivery incentive rule | `name`, `partner_id` / `restaurant_id` / `zone_id`, `dpd_target`, `dpd_period`, `start_date`, `end_date`, `reward_mode`, `payout_mode`, `pdp_targets`, `priority`, `is_active` | 1:N `delivery_rule_scopes`, `incentive_rule_tiers` |
| `delivery_rule_scopes` | Multi-scope rule targeting | `rule_id`, `scope_type`, `scope_id` | N:1 `delivery_rules` |
| `incentive_rules` | Tiered / cumulative plans | `overrides_others`, `priority`, `tiers`, `applies_to` | 1:N `incentive_rule_tiers`, `incentive_rule_scopes` |
| `incentive_rule_tiers` · `incentive_rule_scopes` | tier rows / scope rows | — | N:1 |
| `driver_earnings_daily` | Priced per rider-day | `{driverId}_{earn_date}`, `deliveries`, `incentive_kwd`, `breakdown`, `calculated_at` | N:1 `drivers` |
| `driver_wallet_entries` | Ledger | `driver_id`, `kind`, `amount_kwd`, `ref` | N:1 `drivers` |
| `driver_payouts` · `payout_runs` | Payout batches | `run_id`, `driver_id`, `amount` | N:1 |
| `payroll_clients` | One row per payroll client (Americana, Keeta, …) | `key`, `display_name`, hour values, thresholds, `zone_efficiency_good` / `_low`, `employment_category`, `rules` | 1:N `payroll_client_rules` |
| `payroll_client_rules` | Ordered rule engine rows | `conditions` (jsonb), `all`, `result` | N:1 `payroll_clients` |
| `payroll_zone_metrics` | Previous-month zone efficiency | `{zoneId}_{YYYY-MM}`, `orders`, `rider_days`, `dpd`, `target_dpd`, `efficiency_pct`, `dpd_used`, `target_used`, `category` | N:1 `zones` |
| `payroll_zone_settings` | Global + per-zone thresholds | `target_dpd`, `good`, `low`, per-zone overrides | — |
| `payroll_manual_adjustments` | Append-only hand edits | `driver_id`, `date`, `status`, `hours`, `reason`, `actor_id`, `undo_batch` | N:1 `drivers` |
| `payroll_day_hours` | Loaded hours for a Keeta/Five Guys day | `{driverId}_{YYYY-MM-DD}` | N:1 `drivers` |
| `payroll_order_overrides` | Manual order count per day | `{driverId}_{YYYY-MM-DD}`, `orders` | — |

| Collection | Purpose | Key fields | Relationships |
|---|---|---|---|
| `payroll_rider_statuses` (`roster` doc) | Status label catalog | `rosters`, `customs` | — |
| `payroll_column_config` | Org-wide column labels + per-view visibility | `views`, `labels` | — |
| `payroll_rule_audit_logs` | Rule change audit | `before`, `after`, `actor_id` | — |
| `rider_compliance_daily` | Invalid / rejected / cancelled days | `{driverId}_{day}` | N:1 `drivers` |
| `order_recon_runs` | One applied AM sheet | `status`, `from_date`, `to_date`, `kpi`, `created_at` | 1:N `order_recon_rows` |
| `order_recon_rows` | Melted AM cells | `run_id`, `employee_id`, `work_date`, `excel_orders`, `app_orders`, `difference` | N:1 `order_recon_runs` |
| `fuel_fills` | Fuel ledger | `driver_id`, `vehicle_id`, `litres`, `cost_kwd`, `fuel_type`, `transfer_type` | N:1 |
| `fuel_withdrawn_overrides` | Per-month withdrawn flag | `{driverId}_{vehicleId}_{YYYY-MM}` | — |

## 2.5 EmployeeDesk: requests, e-signature, visits

| Collection | Purpose | Key fields | Relationships |
|---|---|---|---|
| `requests` | RCM case | `request_code` (`RCM-####`), `request_type` (FK → `request_type_definitions.key`), `status`, `driver_id`, `payload` (jsonb), `current_step_order`, `attention_at`, `acknowledged_at`, `sla_due_at`, `closed_at`, `fuel_transfer_type`, `vehicle_id`, `is_confidential` | 1:N `request_approval_steps`, `request_attachments`, `request_clarifications` |
| `request_type_definitions` | Type catalog (8 system + custom) | `key`, `is_system`, `terminal_status`, `requires_ack` | 1:N `request_field_definitions` |
| `request_field_definitions` | Per-type field schema | `type_key`, `key`, `field_type`, `options`, `is_required`, `is_server_required`, `min_value` / `max_value`, `visibility` | N:1 |

| Collection | Purpose | Key fields | Relationships |
|---|---|---|---|
| `request_approval_step_templates` | Chain definition | `type_key`, `step_order`, `role_key`, `sla_minutes`, `breach_action` | 1:N `request_approval_steps` |
| `request_approval_steps` | Live step | `request_id`, `step_order`, `status`, `decided_by`, `actor_display_name`, `sla_due_at` | N:1 `requests` |
| `request_attachments` · `request_clarifications` · `request_comments` · `request_forwards` | Case children | — | N:1 |
| `request_staff_access` · `esign_staff_access` | Per-type sender/receiver grants | `type_key`, `can_view`, `can_decide` | — |
| `esign_templates` · `esign_template_fields` | Template + field map | `kind` (`penalty` \| `loan` \| `payslip` \| `general_doc` \|…), `sheet_html`, `sheet_render_version`, `field_values` | 1:N |
| `esign_categories` | Category catalog incl. `parent_key` | `key`, `screenshot_restricted`, `is_active` | self 1:N |
| `esign_requests` | One signature request | `request_code` (`SIG-####`), `category_key`, `status`, `sent_at`, `viewed_at`, `declined_at`, `declaration_accepted_at`, `document_storage_key`, `signed_document_storage_key`, `employee_snapshot`, `field_values` | 1:N `esign_request_signers`, `esign_request_attachments` |
| `esign_request_signers` | Multi-signer | `request_id`, `driver_id` **xor** `staff_user_id`, `sign_order`, `signed_at` | N:1 |
| `esign_batches` · `esign_batch_rows` | Bulk send (`BAT-####`) | `template_id`, `source_filename`, `total_count`, `progress` | 1:N |
| `esign_drafts` · `esign_audit_events` | Drafts / audit | — | — |
| `visit_branches` | Location | `key`, `name`, `is_default`, `working_dows`, `opening_time`, `closing_time`, `lunch_start` / `lunch_end` | 1:N `visit_departments`, `visit_slots` |
| `visit_departments` | Desk | `key`, `label_en`, `label_ar`, `branch_id` (null = all), `desks_count`, `avg_handling_minutes`, `is_active` | N:1 `visit_branches` |

| Collection | Purpose | Key fields | Relationships |
|---|---|---|---|
| `visit_slots` | Availability | `department_key`, `branch_id`, `weekday`, `start_time`, `end_time`, `capacity` | N:1 |
| `visit_bookings` | Booking (`VIS-#####`) | `booking_code`, `driver_id`, `department_key`, `branch_id`, `slot_id`, `scheduled_date`, `status`, `checked_in_at`, `note`, `note_to_rider`, `vehicle_id` | N:1 |
| `visit_booking_notes` · `visit_blocked_dates` | Admin thread / blackout dates | — | — |

## 2.6 Notifications, performance, fleet service

| Collection | Purpose | Key fields |
|---|---|---|
| `notification_campaigns` | Campaign | `title`, `body`, `status` (`draft` \| `scheduled` \| `sent` \| `failed` \| `partially_failed`), `target_spec`, `audience_snapshot_id`, `screenshot_restricted`, `track_engagement`, `created_at` |
| `notification_templates` | Reusable body + action preset | `action_type`, `fields`, `record_type`, `is_archived` |
| `notification_audience_snapshots` | Frozen audience | `ids[]`, `count`, `spec` |
| `notification_dispatch_runs` · `notification_dispatch_items` | Per-campaign send ledger | `campaign_id`, `driver_id`, `status`, `dismissed_at`, `opened_at` |
| `notification_automations` · `notification_automation_events` | Trigger → send rules | `kind`, `lead_days`, `is_active` |
| `notification_client_events` · `notification_events` · `notification_analytics_daily` · `notification_timeline` | Engagement | `{day}_{campaignId}`, delivery / open / click / failure |
| `notification_device_tokens` · `driver_push_tokens` | FCM registry | `token`, `is_active`, `last_seen_at` |
| `driver_performance_daily` | **Raw numerators and denominators**, never a score | per rider-day component inputs |
| `performance_score_components` | Weighted components (`punctuality`, `duty_ratio`, `on_time`, `speed`, `zone`, `gps`, `conduct`) | `key`, `weight`, `sort_order`, `is_active` |

| Collection | Purpose | Key fields |
|---|---|---|
| `performance_rating_criteria` | Team × criterion | `team_key` , `key` , `weight` |
| `driver_performance_ratings` | `(driver_id, criterion_id, period_month)` | `score` 1–5 |
| `driver_performance_rating_notes` | `(driver_id, team_key, period_month)` | `note` |
| `performance_rating_team_members` | Who may rate which team | `team_key` , `user_id` |
| `vehicle_services` | Workshop ledger (legacy + additive) | `vehicle_id` , `service_type_key` , `odometer_km` , `cost_kwd` , `stage` , `parts[]` |
| `service_types` | Service Type catalog: **468 part rows** + 5 maintenance programs, 12 categories | `key` , `label_en` , `label_ar` , `category` , `tracks_maintenance` |
| `vehicle_odometer_readings` | One reading per service or manual edit | `vehicle_id` , `km` , `source` ( `dashcam` \| `manual` \| `service` ), `previous_km` , `lowered` , `reason` |
| `vehicle_maintenance_schedules` | Rule per vehicle / make / make-model / type | `scope` , `km_interval` , `day_interval` , `alert_km` , `alert_days` |
| `vehicle_maintenance_exclusions` · `_overrides` | Opt-out / hand-set next service | `{vehicleId}_{programKey}` |
| `vehicle_day_distance` | Bike daily distance | `{vehicleId}_{day}` , `total_meters` , `by_driver` , `sessions` |
| `service_maintenance_settings` · `service_odometer_sync` | Singleton `current` docs | provisional intervals / dashcam config |
| `vehicle_accidents` | Accident case | `reference` ( `ACC-####` ), frozen vehicle + driver snapshot, `stage` , `cost_kwd` , `attachment_keys[]` |
| `asset_catalog` · `asset_assignments` | Inventory + issued assets | `category` , `penalty_kwd` , `total_quantity` , `quantity` |
| `custom_field_definitions` | **Admin-defined columns** | `entity_type` ( `driver` \| `partner` \| `restaurant` \| `zone` \| `vehicle` ), `key` , `label` , `field_type` , `options` , `required` , `is_active` , `sort_order` |
| `document_tracking` | Expiry per document | `driver_id` , `doc_type` , `expiry_date` , `lead_days` |
| `wrong_actions` | Conduct incidents | `driver_id` , `action_type` , `severity` ( `low` \| `medium` \| `high` ), `occurred_at` |
| `order_exceptions` | Rider-filed rejected / cancelled order | `driver_id` , `kind` , `source` , `order_id` , `reason` , `proof_keys[]` , `status` , `day` |

| Collection | Purpose | Key fields |
|---|---|---|
| `driver_groups` · `driver_group_members` | Grouping | `member_ids[]` mirrored |
| `source_companies` · `employee_statuses` · `employee_sections` · `nationality_categories` · `bank_codes` · `job_positions` · `sponsorships` | Employee master catalogs | `key` , `label_en` , `label_ar` , `is_active` , `sort_order` |
| `support_threads` · `support_messages` · `appointment_slots` · `appointments` | Support + meetings | — |
| `client_portals` | One portal config per project key | `project_key` , `theme` , `rate_cards` |
| `driver_import_batches` · `driver_import_rows` | Bulk employee import job | `status` , `heartbeat_at` , `pending_store` , `uploaded_by` |
| `page_read_snapshots` · `page_read_generations` | Gzipped read-through cache for scan-lane pages | `packed` , `generation` , `fresh_until` |
| `admin_cost_samples` | Sampled cost telemetry | — |
| `admin_allowlist` | Sign-up allowlist | `email` |

## 2.7 Entity-relationship diagram (core)

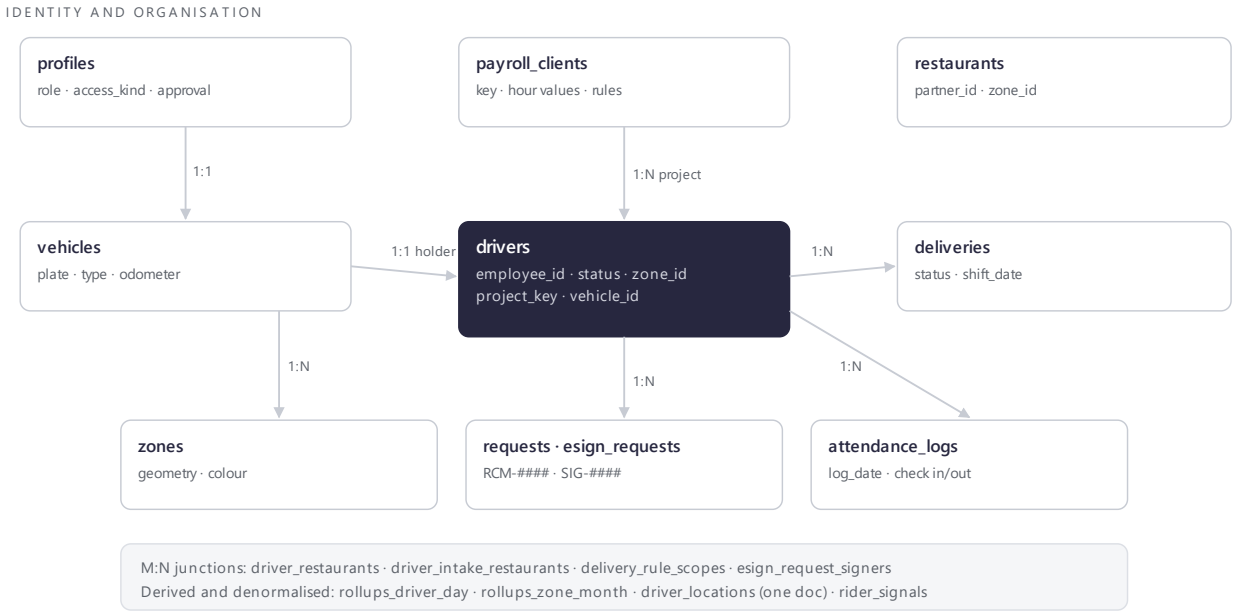



**Figure 2.** The core graph. Everything the console lists hangs off drivers; everything the rider files hangs off drivers too, which is why one identity fix corrects a dozen screens.

# 3 Core Application Functions & Business Logic

## 3.1 Session, RBAC and permission resolution

The chain is deliberately short — a page render costs a fixed number of reads, not one per permission.

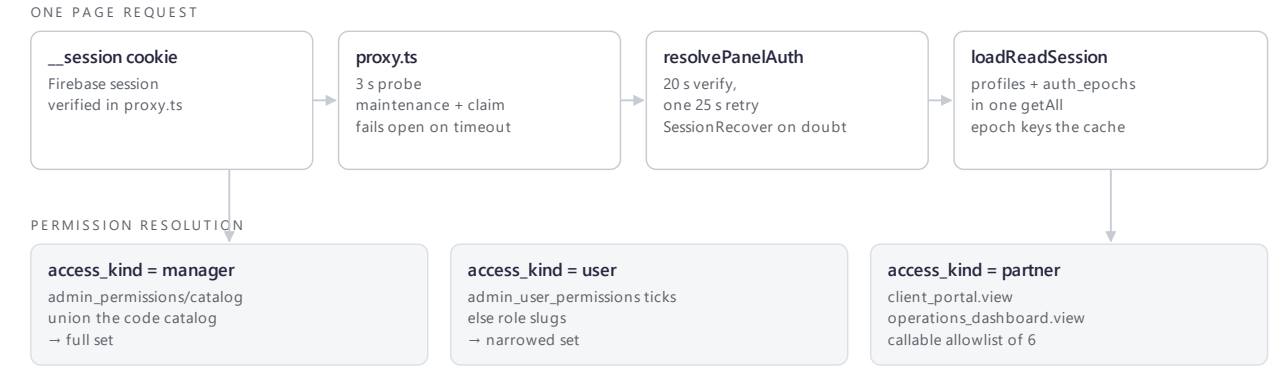



**Figure 3.** Session to permission in one pass. A slow or unreachable Auth backend resolves to SessionRecover — an unproven session is never treated as a signed-out one.

| Function | File | Behaviour |
|---|---|---|
| `panelGate` | `src/lib/auth/staff-entry.ts` | Pure state machine: `login` / `not_authorized` / `claim` / `pending` / `maintenance` / `unavailable` / `ok` |
| `resolvePanelAuth` | `src/lib/auth/require-permission.ts` | Returns `SessionRecover` instead of throwing when auth is unproven — a slow GoTrue is not a logout |
| `requirePermission` / `requireAnyPermission` / `requireSuperAdmin` / `requireNotPartner` | same | Page guards; redirect to `/unauthorized` |
| `loadReadSession` | `src/lib/auth/epoch-session.ts` | `auth_epochs/current` + `profiles/{uid}` in **one** `getAll` ; permission docs re-read against the epoch on a miss |
| `enrichSessionPermissions` | `src/lib/auth/profile-auth.ts` | Manager = catalog ∪ code catalog ( `catalogWithCodeSlugs` ); User = ticks; Partner = 2 fixed slugs |
| `permissionGrantedByTicks` | `src/lib/auth/staff-access.ts` | `*.manage` ↔ create/edit/delete alias; `bulk_delete` **never granted by another tick** |
| `hasPermissionInSet` | `src/lib/auth/permissions.ts` | Super admin always passes |

**The four access levels** on `/settings/roles` ( `src/lib/auth/app-access.ts` ) map onto existing slugs — no parallel permission model:

| Level | Effect |
|---|---|
| No access | No slug |
| Viewer | `*.view` |
| User | create/edit/delete |
| Manager | `*.manage` + `*.bulk_delete` |

**Epoch caches.** `auth_epochs/current` holds counters ( `access` , `settings` , `menu` , `payroll` , …). Every role, tick, catalog, branding and menu write goes through `writeWithEpoch` , which increments the counter in the same transaction; readers key their cache on it. A change therefore lands on the next request, with a 5-minute TTL as the safety net. `clearCatalogSlugsCache()` is the test seam.

**Partner scoping.** `access_kind: 'partner'` is locked to `partner_project_key` ∈ { `keeta` , `americana` , `unassigned` , `five_guys` } and only two slugs. `requireStaff` refuses every callable outside `PARTNER_CALLABLE_ALLOWLIST` (6 names), and masked fields (phone, bank, salary, leave reason) are dropped server-side in `functions/src/rpcs/*` — never in the browser.

## 3.2 Catalog synchronisation (Employees ↔ HR Payroll ↔ Profiles)

The client's 5.2 ask — "a change made once reflects everywhere" — is implemented as **one catalog, many readers**, not as a sync job.

| Catalog | Collection | Written by | Read by |
|---|---|---|---|
| Partners / Clients | `partners` | `/partners` , OperationsHub | Employees form (Client dropdown), Payroll Client Name, Client Portal, Incentive rule scope |
| Restaurants | `restaurants` | `/restaurants` | Employees (assignment), Payroll (Restaurant column), Performance (restaurant DPD), Delivery proximity gate |
| Zones | `zones` | `/zones` | Employees, Live Tracking filter, Payroll Zone Category, Order recon |
| Vehicle types | `vehicle_types` | `/settings/vehicle-types` | Vehicles form, Employees Vehicle column, Fuel limit default |
| Employee statuses | `employee_statuses` | `/settings/employee-statuses` | Employees form, HR Payroll Status paint, Payroll status pill |
| Employee sections | `employee_sections` | `/settings/employee-sections` | Employees form, approval gate ( `is_operational` ) |
| Nationality categories | `nationality_categories` | `/settings/nationality-categories` | Employees form, HR Payroll Nationality, eSign snapshot |
| Sponsorships | `sponsorships` | `/settings/sponsorships` | Employees form, HR Payroll Sponsor, eSign snapshot |
| Service types | `service_types` | `/settings/service-types` | Fleet Service records, maintenance programs |
| Payroll clients | `payroll_clients` | `/payroll/settings` | Combined Payroll, HR Payroll, Client name column |

Cache invalidation is by **query-key factory** ( `src/lib/query/query-keys.ts` ) plus `revalidatePath` on the affected routes. There is no polling sweep: a save writes the catalog and invalidates the readers.

**The sync bug this fixed.** A new partner added in OperationsHub did not appear in the Employee Client dropdown because the dropdown read a module-local list. It now reads `partners` through the same server read the OperationsHub page uses. `source_companies` (Company) and `sponsorships` follow the same rule: every active row is listed, and a stored-but-inactive value is preserved rather than snapped to a default.

## 3.3 Payroll engines

Two engines, deliberately separate:

### A. Payroll Hub month grid ( `src/features/payroll/` )

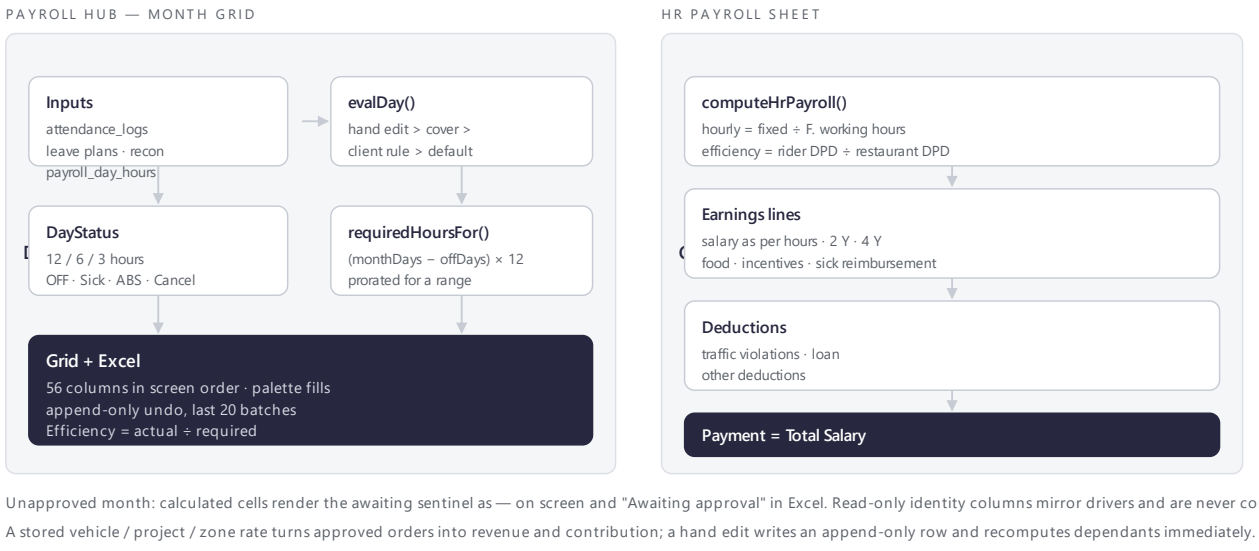



**Figure 4.** Two engines, deliberately separate: the month grid decides what a day was, HR Payroll prices the month. Neither recalculates the other.

| Concern | Rule | File |
|---|---|---|
| Work day unit | `PAYROLL_DAY_HOURS = 12` | `payroll-formulas.ts` |
| Default OFF | `PAYROLL_DEFAULT_OFF_DAYS = 2` per month | same |
| Required hours | `(monthDays − offDays) × 12` , prorated for a partial range | `requiredHoursFor` , `requiredHoursForRange` |
| Day precedence | hand adjustment > Operations-approved cover > client rule > client default > legacy fallback | `payroll-rules-engine.ts` |
| Cell label | numeric ( `12` , `6` , `3` , `4.3` ) for hours; words for `OFF` / `Sick` / `ABS` / `Cancel` | `dayGridLabel` |
| Zone efficiency | previous **completed** month: `orders ÷ rider_days` vs fleet average; Good ≥ 110, Average ≥ 70, Low < 70 | `payroll_zone_metrics` |

| Concern | Rule | File |
|---|---|---|
| Undo | append-only batches, `reason: 'undo'`, last 20 | `payroll_manual_adjustments` |
| Export | 56-column Excel matching the screen exactly, palette fills included | `combined-xlsx.ts` |

### B. HR Payroll sheet ( `src/features/payroll/hr-payroll-formulas.ts` )

Client-specified field rules (review §3.4), implemented verbatim in `computeHrPayroll`:

| Field | Rule |
|---|---|
| Payment | `= Total Salary` |
| Hourly Rate as per F. Working Days | `Fixed Salary ÷ F. Working Hours` |
| Efficiency | `Rider DPD ÷ Restaurant DPD` (0 when the denominator is 0) |
| Salary as per Hours | `Hourly Rate × Actual Working Hours` |
| 2 Years Increment | `2-Year Applicable ÷ F. Working Days × Working Days` |
| 4 Years Increment | `4-Year Applicable ÷ F. Working Days × Working Days` |
| Sick Leave Reimbursement | `Work Permit Salary ÷ 26 × Sick Leave Days` |
| Total Salary | `Salary as per Hours + 2Y + 4Y + Incentives + Food Allowance + Sick Reimbursement + Pending Salary + Total Allowances − Traffic Violations − Loan Deduction − Total Deductions` |
| Below-contract highlight | `hrSalaryShortfall()` returns a positive gap → row painted red when `Total Salary < Work Permit Salary` |
| Unapproved month | Cells that depend on Ops approval render `HR_AWAITING` ( `\u0000awaiting` ) → `—` on screen, "Awaiting approval" in Excel |

**Important separation:** "From employees" columns (Nationality, Status, Work Start, Job End) are **read-only reflections** of `drivers`; only the Configuration and Calculated blocks are computed or editable. A hand edit is written through `adminSaveHrPayrollCells` and recomputes the dependent cells immediately.

### C. Per-delivery incentive ( `functions/src/core/incentive.ts` )

```
target  = restaurant rule → zone rule → partner rule → rider's own zone/partner
bands   = tier rates applied to consecutive bands above the target
gate    = compute_incentive_amount honours the DPD target for
          EVERY reward_mode / payout_mode / overrides_others
history = band math starts at app_settings.incentive_band_math_from
```

## 3.4 Inline editing, optimistic UI and read-through caching

| Mechanism | Where | Behaviour |
|---|---|---|
| Optimistic mutation | `updateDeliveryStatus`, payroll cell edits, notification templates | Patch the cache first, then commit; a failed side-effect (earnings recalc) never discards the write |
| Append-only edit log | `payroll_manual_adjustments` | Every hand edit stores `before`, `after`, `reason`, `actor_id`; undo is a new row, never an update |
| Autosave | Payroll cell edit, HR cell edit | Debounced; a toast only on add/delete/reset |
| Read-through snapshot | `functions/src/router/read-snapshot.ts` | Scan-lane results are gzipped into `page_read_snapshots`, keyed by `fn |
| Epoch-keyed caches | `epoch-cache.ts`, `config-cache.ts` | Bounded LRU per isolate; invalidated by counter, not by TTL alone |
| Previous data held | `placeholderData: keepPreviousData` on every list | Changing a filter does not blank the table |

## 3.5 Global search, filters and clickable KPI cards

| Capability | Implementation |
|---|---|
| **Row search** | `fieldSearchMatch(row, query, accessors)` — one matcher across list pages; surname, code, restaurant, `EMP` digit run |
| **Per-column filters** | `ComparisonColumnFilters` / `PayrollColumnsMenu` / `TableToolbar` — text contains, value list with explicit **All**, numeric **range** (min/max), date range |
| **Group by** | `groupRows()` + `useGroupedListView()` — groups start closed, header shows value + count, header checkbox selects the whole group across its pages |
| **Clickable KPI cards** | `KpiCard` accepts `onClick` + `selected`; the tile applies the filter and the table, charts and sibling tiles re-derive from the same selector. Client portal uses `riderCountsByWidget` / `filterRidersByWidget` so a tile and the table it opens cannot disagree |
| **Table toolbar** | `TableToolbar` = Search + Group by + List/Grid + Columns, all URL-backed where sensible and persisted per user in `admin_ui_preferences` |
| **Numeric-only guard** | MG ID / Civil ID filters strip non-digits before the predicate (`sanitizeNumericFilter`) |

## 3.6 Domain gates that carry business rules

| Gate | Rule | File |
|---|---|---|
| Delivery proximity | In range when inside the assigned **zone polygon or any assigned store range**; store pin wins over polygon | `DeliveryProximityService.evaluate`, `isWithinDeliveryRange` |
| `zoneDescribesWorkArea` | A zone holding none of the rider's stores is **withheld** from the app, so the store range decides | `functions/src/rpcs/driver-deliveries.ts` |

| Gate | Rule | File |
|---|---|---|
| Zone idle timeout | 45 min outside **zone or store range** → auto checkout; 20 min return grace after a delivery | `zone_monitor_provider.dart`, `admin_run_attendance_auto_checkout` |
| Pickup order ID | Digits only, 1–32; unique **per store per Kuwait day**, not fleet-wide | `driverCreatePickup` |
| Stale pickup | `app_settings.pickup_auto_cancel_hours` (default 6) → cron cancels and notifies | `/api/cron/expire-stale-pickups` |
| Driver level D | Month-to-date: `activeDays ≥ 3` and `verified < activeDays × target` (fallback 25) | `functions/src/rpcs/driver-levels.ts` |
| Level D override | Admin override writes a notification with deep link `musallam:///profile/performance-level` | same |
| Accident case | Frozen vehicle + driver snapshot so a later plate change cannot rewrite history | `vehicleAccidents` |
| Maintenance program | One row per **vehicle × program**; no approved rule ⇒ `no_rule`, not a guess from defaults | `service-maintenance-model.ts` |
| Odometer | Only ever moves **up** on unattended paths; a lower reading needs a written reason | `odometerDecision`, `recordManualOdometer` |
| E-sign stamp | Named slot per signer, never "last page bottom-right"; `sheet_render_version` gates measured boxes | `esign-signature-slots.ts` |
| Request SLA | `sla_minutes` per step; the hourly sweep **raises attention, never decides** | `admin_run_request_sla_sweep` |

# 4 Cloud Functions, Serverless Endpoints & Background Jobs

## 4.1 Callable routing — three lanes, one cold start

All callables live in **one** Firebase functions package and are reached through three exported `onCall` routers plus the legacy names. `functions/src/router/lanes.ts` is the single source of truth, shared by the functions package and the browser bundle.

| Lane | Export | Region | CPU / Mem | Concurrency | Instances | Timeout | Serves |
|---|---|---|---|---|---|---|---|
| **light** | `adminApi` | `me-central1` | 1 vCPU / 2 GiB | 8 | 0–8 | 60 s | Every admin callable not on scan/heavy |
| **scan** | `adminApiScan` | `me-central1` | 1 vCPU / 4 GiB | 2 | 0–6 | **180 s** | 19 unbounded readers (lists, payroll snapshots, attendance, fleet snapshot, order comparison…) |
| **heavy** | `adminApiHeavy` | `me-central1` | 1 vCPU / 4 GiB | 1 | 0–3 | 300 s | Purges (17), earnings recalculations, e-sign compose, import chunk claim, level sweep |

`LANE_CLIENT_TIMEOUT_MS` keeps the browser abort ≥ the server timeout: `light 60 s`, `scan 180 s`, `heavy 300 s`. `laneOf(name)` decides from the registry, so a new callable cannot be shipped on the wrong lane; `assertPartition()` fails the build if a name is on two lanes or missing from the exports.

**Scan-lane membership (19):** `adminPerformanceOpsSnapshot`, `adminOrderComparisonSnapshot`, `adminOrderReconCompare`, `adminDeliveriesStatusCounts`, `adminDeliveriesCountsByFilters`, `adminListDriversPage`, `adminDriversFilterValues`, `adminLiveFleetSnapshot`, `adminPayrollRuleSnapshot`, `adminPayrollMonthSnapshot`, `adminHrPayrollSnapshot`, `adminHrAppHours`, `adminListAttendanceDaily`, `adminAttendanceKpis`, `adminIncentiveDailyReport`, `adminEmployeesDashboard`, `adminOperationsDashboard`, `adminClientPortalDashboard`, `adminListDriverLevels`.

## 4.2 Callable inventory — 319 exports across 73 modules

| Module file | Exports | Representative names |
|---|---|---|
| `drivers.ts` | 7 | `adminListDriversPage`, `adminDriversFilterValues`, `adminUpsertDriver`, `adminArchiveDriver` |
| `drivers-admin.ts` | 8 | `adminApproveDriver`, `adminCreateDriverIntake`, `adminRestoreDriverIntake` |
| `driver-home.ts` | 3 | `driverGetHomeDashboard`, `driverSetDutyState`, `driverCheckInContext` |
| `driver-deliveries.ts` | 5 | `driverCreatePickup`, `driverCompleteDelivery`, `driverCancelDelivery`, `driverGetDeliveryProximityContext` |
| `driver-location.ts` | 2 | `driverReportLocation`, `driverClearLiveLocation` |

| Module file | Exports | Representative names |
|---|---|---|
| `driver-earnings.ts` | 4 | `driverGetEarningsSummary`, `driverGetExtraEarnings`, `driverGetEarningsDetail` |
| `driver-levels.ts` | 4 | `driverGetMyLevel`, `adminListDriverLevels`, `adminSetDriverLevelOverride`, `adminEvaluateDriverLevelsSweep` |
| `driver-requests.ts` | 7 | `driverCreateRequest`, `driverClarifyRequest`, `driverAcknowledgeRequest`, `driverRespondReschedule` |
| `driver-esign.ts` | 8 | `driverListEsignRequests`, `driverGetEsignRequest`, `driverSubmitEsignature`, `driverDeclineEsignature`, `driverMarkEsignViewed`, `driverAddEsignAttachment` |
| `driver-visits.ts` | 3 | `driverListVisitSlots`, `driverBookVisit`, `driverCancelVisit` |
| `driver-vehicle.ts` | 2 | `driverGetAssignedVehicle`, `driverReportFuelFill` |
| `driver-devices.ts` | 6 | `driverReportDeviceMeta`, `driverRegisterDevice`, `driverReleaseDeviceSession` |
| `driver-push-tokens.ts` | 2 | `driverUpsertPushToken`, `driverDeactivatePushToken` |
| `driver-telemetry.ts` | 1 | `driverIngestTelemetry` (≤100 events, `event_id` idempotency) |
| `driver-lists.ts` | 10 | `driverListMyDeliveries`, `driverListNotifications`, `driverListTenureOptions` |
| `driver-shift.ts` / `driver-attendance-get.ts` / `driver-pay.ts` / `driver-security.ts` / `driver-profile.ts` / `driver-auth.ts` | 2 / 1 / 2 / 1 / 3 / 3 | `driverSubmitDailyShift`, `driverGetTodayShift`, `driverGetAttendance`, `driverGetPayCalculator`, `driverLogSecurityEvent`, `driverUpdateAvatar`, login/passcode |
| `accidents.ts` | 2 | `driverReportAccident`, `driverListMyAccidents` |
| `payroll.ts` | 1 | `adminPayrollRuleSnapshot` |
| `payroll-more.ts` | 7 | `adminPayrollMonthSnapshot`, `adminSavePayrollIdentity` |
| `payroll-edits.ts` | 6 | `adminApplyPayrollAdjustments`, `adminSetPayrollColumnConfig` |
| `payroll-clients.ts` | 9 | `adminListPayrollClients`, `adminSavePayrollClient`, `adminDeletePayrollClient` |
| `hr-payroll.ts` | 3 | `adminHrPayrollSnapshot`, `adminSaveHrPayrollCells` |
| `attendance-ops.ts` | 7 | `adminListAttendanceDaily`, `adminAttendanceKpis`, `adminCorrectAttendance`, `adminRunAttendanceAutoCheckout` |
| `attendance.ts` | 2 | `adminListAttendanceRequests` |
| `requests-core.ts` | 6 | `adminListRequests`, `adminGetRequest`, `adminCreateRequest` |
| `requests-workflow.ts` | 9 | `adminDecideRequest`, `adminForwardRequest`, `adminRunRequestSlaSweep`, `adminAutoCloseRequests` |

| Module file | Exports | Representative names |
|---|---|---|
| `esign-requests.ts` | 12 | `adminListEsignRequests`, `adminGetEsignRequest`, `adminCreateEsignRequest`, `adminRemindEsignRequests`, `adminExpireEsignRequests` |
| `esign-batches.ts` | 11 | `adminCreateEsignBatch`, `adminProcessEsignBatchChunk`, `adminRetryEsignBatchRow` |
| `esign-signers.ts` | 8 | `adminListEsignSigners`, `adminAddEsignSigner`, `adminReorderEsignSigners` |
| `esign-compose.ts` | 1 | `esignComposeSignedDocument` (heavy) |
| `visits.ts` / `visits-shared.ts` | 5 | `adminListVisits`, `adminUpdateVisitStatus`, `adminRescheduleVisit`, `adminSetVisitNoteToRider` |
| `operations-dashboard.ts` | 4 | `adminOperationsDashboard`, `adminClientPortalDashboard`, `adminListLeavePlans` |
| `performance-list.ts` | 2 | `adminListDriverPerformance`, `adminDriverPerformanceDaily` |
| `performance-extra.ts` | 6 | `adminPerformanceOpsSnapshot`, `adminDpdEfficiencySnapshot`, `adminPerformanceTrend`, `adminPerformanceOpsBounds` |
| `performance-ratings.ts` | 6 | `adminListDriverPerformanceRatings`, `adminUpsertDriverPerformanceRating`, `adminSetPerformanceTeamMember` |
| `rules.ts` | 11 | `adminListDeliveryRules`, `adminUpsertDeliveryRule`, `adminListIncentiveRules`, `adminUpsertIncentiveRule` |
| `catalogs.ts` | 21 | `adminUpsertVehicleType`, `adminUpsertVehicleCompany`, `adminUpsertEmployeeCatalog`, `adminUpsertRestaurant`, `adminUpsertPartner`, `adminUpsertZone`, `adminUpsertServiceType` |
| `recon-reports.ts` | 5 | `adminOrderComparisonSnapshot`, `adminOrderReconCompare`, `reconcileDeliveryVerification`, `reportDeliveryOrders` |
| `notifications.ts` | 2 | `adminCreateCampaign`, `adminSendCampaign` |
| `notification-audience.ts` | 4 | `compileNotificationAudience`, `estimateNotificationAudience`, `resolveImportDriverIds` |
| `fleet.ts` | 2 | `adminLiveFleetSnapshot`, `adminGetDriverDayRoute` |
| `fleet-worker-http.ts` | 3 | `workerFleetRead`, `adminIngestDriverPositions`, `adminRecordFleetEvents` (onRequest + `X-Worker-Secret`) |
| `day-route.ts` | 1 | `adminGetDriverDayRoute` |
| `purge.ts` / `purge-all.ts` / `purge-entities.ts` | 5 / 12 / — | 24 entities, `_admin_purge_require(slug)` gate, FK release before delete |
| `retention.ts` | 5 | `cleanupDriverLocationEvents`, `cleanupDriverOperationEvents`, `cleanupDriverTelemetryEvents`, `cleanupStaleDriverLocations` |

| Module file | Exports | Representative names |
|---|---|---|
| `service-odometer-sync.ts` / `service-distance-sweep.ts` | 1 / 1 | scheduled, see §4.4 |
| others | — | `fuel.ts`, `payouts.ts`, `earnings.ts`, `driver-list-plan.ts`, `misc-admin.ts`, `sync-staff-claims.ts`, `home-snapshot-repair.ts`, `leave-plans.ts`, `loan-tenure.ts`, `order-exceptions.ts`, `employee-dashboard`, `deliveries-counts/status`, `check-in-gate`, `delivery-context*`, `bites-*` |

## 4.3 Next.js API routes — 53 `route.ts` files

| Group | Routes | Method | Auth | Purpose |
|---|---|---|---|---|
| **Cron** (13) | see §4.5 | GET/POST | `CRON_SECRET` | Scheduled sweeps |
| **Order recon** | `/api/order-recon/{preview,commit,comparison,undo,redo}` | POST | session + `order_recon.*` | 300 s budget; Excel melt, compare, replay |
| **Driver uploads** | `/api/driver-uploads/{presign,confirm,proxy,read,mine}` | POST/GET | rider session | R2 presign → PUT → confirm; proxy streams when S3 signing is unavailable |
| **eSign** | `/api/esign/{document-download,batch-zip,render-bench}` | GET/POST | `esign.sign` or `requests.manage` | Same-origin streaming so an expired JWT cannot modal an error |
| **Deliveries** | `/api/deliveries/proof-download`, `/api/deliveries/[id]/gps` | GET | `deliveries.view` | Attachment download; day route |
| **Driver app** | `/api/driver-app/{active-release,firebase-config,notifications,notification-media}` | GET/POST | anon + rider token | Force-update gate, config, inbox |
| **Admin ops** | `/api/admin/{app-releases/*,driver-documents/*,locales/*,translations/*}` | GET/POST | `settings.manage` | Release upload, doc upload, locale editor |
| **Assistant** | `/api/assistant/chat` | POST | `assistant.view` | AI Gateway ( `openai/gpt-4o` ), 8 steps, read-only tools |
| **Misc** | `/api/build-id`, `/api/storage/signed-url`, `/api/launcher/weather`, `/api/live-tracking-v2/token`, `/api/menu-icons/[file]`, `/api/notifications/events`, `/api/drivers/import-template.xlsx` | GET/POST | mixed | Version gate, signed URLs, weather (Open-Meteo, 30 min), V2 socket ticket, uploaded menu icons, engagement beacon, template download |

## 4.4 Cloud Functions — scheduled (`onSchedule`, 5)

| Function | Schedule (Kuwait) | Purpose |
|---|---|---|
| `serviceOdometerSync` | `0 0,6,12,18 * * *` | Dashcam odometer pull; reads `service_odometer_sync/current`; secret `SERVICE_ODOMETER_APP_SECRET`; never rewinds a reading |
| `serviceVehicleDistanceSweep` | `45 23 * * *` | Nightly bike day-distance for riders who never clocked out |
| `deliveryProximitySweep` | periodic | Rebuilds the shared delivery-context document (fleet settings, zones, restaurants + geofences) |
| `homeSnapshotRepair` | periodic | Repairs `driver_home_daily` rows |
| `bitesMirrorSweep` | periodic | Bites order mirror reconciliation |

## 4.5 Vercel cron jobs (`vercel.json`, 13)

| Path | Cron | Purpose |
|---|---|---|
| `/api/cron/notifications-dispatch` | `*/5 * * * *` | FCM dispatch for scheduled campaigns |
| `/api/cron/notifications-automations` | `*/5 * * * *` | Lead-day automations (document expiry, visits) |
| `/api/cron/attendance-auto-checkout` | `*/5 * * * *` | 45-min zone exit, shift end, orphaned duty |
| `/api/cron/dpd-shift-notices` | `*/5 * * * *` | 30-min warning, target reached, shift summary |
| `/api/cron/esign-batch-drain` | `*/5 * * * *` | Finishes a bulk e-sign batch after the tab closes |
| `/api/cron/locations-cleanup` | `*/2 * * * *` | Retire stale `driver_locations` |
| `/api/cron/expire-stale-pickups` | `*/15 * * * *` | Cancel a pickup stuck `in_transit` |
| `/api/cron/roster-alerts` | `*/15 * * * *` | No-login, late, early-drop, overtime |
| `/api/cron/requests-sweep` | `0 * * * *` | SLA breach attention + auto-close |
| `/api/cron/storage-uploads-cleanup` | `0 4 * * *` | Orphaned uploads |
| `/api/cron/driver-ops-retention` | `20 1 * * *` | 30-day operation events |
| `/api/cron/driver-telemetry-retention` | `40 1 * * *` | 14-day telemetry |
| `/api/cron/performance-rollup` | `0 2 * * *` | Rewrite trailing `driver_performance_daily` window |

## 4.6 Cloudflare Worker `dpd-live`

| Piece | File | Role |
|---|---|---|
| Worker entry | `infra/workers/dpd-live/src/index.ts` | Routes `/ingest`, `/ingest/ws`, `/ws`, `/latest`, `/fleet-stats`, `/history`, `/history/days`, `/archive/*` |
| **`FleetRoom` Durable Object** | `fleet-room.ts` | One room = whole fleet. Holds roster, zones, shifts, thresholds and a 10-minute trail buffer in memory; batches a durable flush every 10 s through `adminIngestDriverPositions` |
| Interest management | `fleet-interest.ts` | Viewport + filter culling before serialise — 500 riders affordable |
| Rules | `fleet-rules.ts` | Shared with `functions/src/rpcs/fleet-ingest-rules.ts`; hysteresis + grouped cooldowns for Class B events |
| GPS archive | `gps-archive.ts` | SQLite + R2 `dpd-gps-prod`; minute files compacted hourly per rider |
| Auth | `auth.ts` | Firebase JWKS verification, tri-state (`ok` / `rejected` / `unavailable`), 5-min negative cache only on a **definitive** rejection |

**Read-cost posture.** `ROSTER_TTL_MS` 10 min; a rider not on the roster is read **individually** (`workerFleetRead { op: "driver_snapshot" }`, ~12 reads) with a 5-minute negative cache, instead of a whole-fleet snapshot per batch. `FLEET_BASE_CACHE_MS` 120 s. `SNAPSHOT_POLL_MS` 60 s on the client.

## 4.7 Webhooks and outward integrations

| Integration | Direction | Transport | Notes |
|---|---|---|---|
| FCM | outbound | Admin SDK `sendEach` | Batched 500/topic; token per driver; stale token picks `max(last_seen_at)` |
| BDashcam / VSS GPS MEA | outbound (blocked) | HTTPS | `/sys/login` + `/position/realtime/vehicleGps`; **gateway rejects the login from our network — awaiting vendor allowlist**, config `enabled:false` |
| Open-Meteo | outbound | HTTPS | `/api/launcher/weather`, 30 min cache |
| AI Gateway | outbound | HTTPS | `/api/assistant/chat` only |
| Cloudflare `dpd-live` | inbound | `X-Worker-Secret` | `workerFleetRead`, `adminIngestDriverPositions`, `adminRecordFleetEvents` |
| Bites | both | `BITES_BRIDGE_SECRET` | `bitesFleetSocketTicket`, `bitesMirrorSweep` |

## 4.8 Audit and observability

| Concern | Implementation |
|---|---|
| Staff audit | `logAdminMutation()` on every write → `admin_activity_logs` with `before` / `after` and `page_path`; driver-document **view and download** are logged too |

| Concern | Implementation |
| --- | --- |
| Rider RPC audit | `log_driver_operation` in-transaction; `log_driver_operation_autonomous` over `dblink` for RPCs that raise (so a refused pickup is still recorded) |
| Callable logs | `writeRouterLog(lane, fn, inflight, ms, snapshot)` → single-line JSON per call |
| Error tracking | Sentry client + functions; `SentryUserSync` binds the staff id |
| Cost telemetry | `admin_cost_samples`, `scripts/admin-cost-audit.mjs` |

# 5 Complete Feature Matrix

| Module | User-facing features | Inline editing | Real-time / sync | Reporting & export |
|---|---|---|---|---|
| **Dashboard / Launcher** | Dark launcher grid, live Kuwait weather, date + clock, badges, ⌘K search | — | Weather 30 min; badges on poll | — |
| **Control Tower** | Cross-module KPI snapshot | — | 60 s poll | — |
| **Live Tracking V1** | Map pins by status, driver popup with address, recent orders, filters, fullscreen | — | 15–30 s poll + `driver_locations` realtime | Day route, CSV |
| **Live Tracking V2** | 1 Hz deck.gl vehicle sprites, 10-min trails, playback scrubber, floating rail + insights, collapsible overlays | — | WebSocket 4 Hz → 1 Hz poll fallback; interpolated on client | Excel |
| **Employees** | List / card view, 40+ columns, tabbed detail (Private info, Documents, Attendance, Assets, Earnings, Loan, Performance, Diagnostics), approve / verify, archive, delete, bulk import + export, custom columns | Cell edit, quick-edit, per-column filters, `custom_fields` | Catalog-driven dropdowns; badge poll | XLSX import (parity-tested) + filtered export |
| **Employee Groups** | Create, member add/remove, Excel import with mismatch rejection | Member chips | — | CSV template |
| **Fleet — Vehicles** | KPI strip (Unassigned/Bike/Car/Active/Replacement), 18 columns, identity header, Assign rider, Assets tab, 5 ledgers | Per-column filters, bulk assign / status / company | Assignment reflected on Employees + Payroll | Sample XLSX + export + import batches with undo/redo |
| **Fleet — Fuel** | Daily/Weekly/Monthly/Custom (≤31 Kuwait days), >90% red, fuel + refund requests, per-driver page | Withdrawn override per rider/vehicle/month | Request queue polls | Weekly/monthly export |
| **Fleet — Service** | Records (5-stage workflow), Cost analysis (KPI, monthly Contract/Service chart, 4 breakdowns), Maintenance (vehicle × program), Bikes page | Odometer dialog, stage change, baseline import | Program rules resolve live | Styled Excel per tab, baseline + bike import |
| **Fleet — Accident** | Case list + 5-stage, frozen vehicle/driver snapshot, costed repair, attachments, detail page, rider-filed cases | Stage, cost, documents | Rider report → inbox | Excel |
| **Fleet — Handover / RJ** | Handover ledger, RJ (rejected-request) list | Record dialog | — | Export |

| Module | User-facing features | Inline editing | Real-time / sync | Reporting & export |
|---|---|---|---|---|
| **Deliveries** | Filters (status, zone, partner, date, flags), bulk verify/reject, detail sheet with proof, GPS trail, Orders Report, reconciliation compare | Status change (optimistic) | Realtime on new rows; 60 s counts | Orders Report matrix XLSX, recon 6-sheet report |
| **Order Exceptions** | Rider-filed rejected/cancelled list, reason + proof count, review dialog with signed URLs | Review approve/reject | Rider push on decision | Summary + XLSX |
| **Payroll Hub — Summary** | 5 KPI cards, status row with counts, 15-row grid, column picker | Identity cells, status pill (custom label) | Snapshot 5 min; epoch-busted on write | Per-tab Excel, CSV |
| **Payroll — Combined** | Spreadsheet grid: 7 identity cols, 31 day cols, ordered summary; drag select, Ctrl+C/V/D, fill handle, paste tiling | Every cell; reason required for hand edit; undo/redo | Optimistic + append-only log | 56-column Excel identical to screen |
| **Payroll — Attendance & Orders** | Dual sub-columns (Payroll / Orders) per day, order override input | Day cell + order count | Recon/recon-override wins | CSV |
| **Payroll — Application Attendance** | Per-day app usage, active days, average hours | — | — | CSV |
| **HR Payroll** | 50 columns in client order, Summary (4 cards + company table), Payroll, Configuration | Configuration + HR cells; calculated block read-only | Awaiting-approval state per month | Excel + import of allowances/deductions |
| **Payroll Settings** | Client criteria, hour values, rule builder, "try the rules", zone efficiency table with overrides, change log | Rules, thresholds, zone overrides | Autosave debounce | — |
| **Attendance** | Live presence, logs, exceptions, analytics, driver explore sheet, correction | Correct check-in/out and status | 60 s; realtime on `attendance_logs` | CSV |
| **Shift Attendance / Roster** | Setup, weekly Excel upload, schedule changes, alerts (ack with outcome), report | Ack outcome | 15-min alert cron | Excel with scheduled/worked hours |
| **Operations Hub** | Hub tiles + two-panel Figma shell | — | — | — |
| **Notifications** | Campaign list with audience counts, 4-step create wizard, templates, automations, history, analytics, per-recipient drill-down | Template body, action fields | Dispatch every 5 min | Engagement report, export |

| Module | User-facing features | Inline editing | Real-time / sync | Reporting & export |
|---|---|---|---|---|
| **Roles & Permissions** | 3-pane access editor (280 / 1fr / 300), Staff rail with dept chips + counts, 4 access levels, sub-view chips, launcher preview, Manager↔User with full-catalog seed, role select, copy access | Ticks, role, level | Save bumps `auth_epochs.access` | — |
| **Menu Editor** | Per-role sidebar tree, drag, hide, rename, reset | Tree edits | Saved config is the sidebar | — |
| **Languages** | Locale table + translation editor writing `src/messages/*.json` | Per-key editor | — | Parity guard via `test:i18n` |
| **Employee Desk — Outgoing** | Template tiles (8 categories + penalty sub-menu), bulk send, tracked sends with progress, admin tools | — | Tracker refreshes on resume | Batch ZIP, unsigned PDF |
| **Employee Desk — Incoming** | 8 queues with counts, needs-action list, filters, sort, upload, forward/escalate, confidential mask | Decision, forward, comment | Push on decision | Per-type queue export |
| **eSign** | Template library + builder, send / bulk (Excel), To Sign inbox, batch detail, remind (cooldown), decline, signed copy, signature proof | Field map, signer order, sheet HTML | Drain cron finishes a closed tab | A4 PDF, signed PDF, signature PNG |
| **Visit Slots** | Hub by department, All visits with header filters, Calendar Day/Week/Month/Year, Reception check-in, Slots, Departments & desks, Branches, Reports | Status, reschedule, note to rider | KPI windows server-side | Excel + QR ticket |
| **Performance** | 5 SOP tabs, 8 slicers, Client theme (Americana blue / Keeta gold), Order Status slicer, sortable + filterable rider table, 60 s refresh | — | 60 s snapshot | Per-chart CSV, rider XLSX |
| **Performance Settings** | Component weights with 3-way preview (old / saved / new), team membership | Weights, members | Save re-scores the fleet | — |
| **Client Portal V2** | Partner-scoped dashboard, 6 clickable KPI tiles → rider table with pager, leave chart, joiners, employee list with documents | — | 2-min snapshot | CSV, document download |
| **Partner Preview** | Super admin "view as Keeta / Americana" with exit banner | — | — | — |
| **Settings — Branding / App** | Name, logo, font, maintenance, driver-app maintenance, force-update minimum version | Forms | Epoch-busted | — |
| **Settings — Catalogs (13)** | Vehicle types/uses/companies, employee statuses/sections, nationality, bank codes, job positions, sponsorships, source companies, service types, payroll clients | Row edit; some bulk import | Every form reads the catalog live | Sample XLSX where applicable |
| **Settings — Custom columns** | **Super Admin only.** Add columns per entity (Employees, Clients, Restaurants, Zones, Vehicles); a `select` / `checkbox` / `date` field also becomes a slicer | Definition editor | Definitions invalidate every list read | — |

| Module | User-facing features | Inline editing | Real-time / sync | Reporting & export |
|---|---|---|---|---|
| **Settings — Data cleanup** | Clear all per module (typed confirmation), Clear by filter, blockers shown first | — | — | Storage key manifest |
| **Driver Devices** | Build / phone / profile / last seen / severity, Force update, notify | Bulk force update | Last-seen poll | XLSX |
| **Document Expiry** | Expired / 1 week / 1 month / 3 months, per-doc lead days | Expiry date, lead days | Automations notify | CSV |
| **Compliance** | Invalid / rejected / cancelled day upload, rider-day ledger | Upload | — | — |
| **Wrong Actions** | Incident list, detail, KPIs by severity, weight feeds Performance | Create / edit | — | CSV |
| **Earnings / Payouts** | Daily earnings, wallet ledger, payout runs + detail | Adjust, recalc range | — | CSV |
| **Support** | Threads, messages, SOS, appointments | Reply | Poll | — |
| **Staff Assistant** | Chat over existing reports (EN + AR, RTL), 8-step tool budget, read-only | — | — | Excel from the same numbers |
| **System** | Cloudflare R2, storage, activity logs, access requests, data cleanup, menu editor, languages | — | — | Log export |

## 5.1 Cross-cutting capabilities present on every list module

| Capability | Mechanism |
|---|---|
| Column order, visibility, sort, group, view | `TableToolbar` + `useTablePrefs` → `admin_ui_preferences` (per user) with role defaults |
| Search | `fieldSearchMatch` ; ≥4-char prefix on indexed fields, full scan fallback while the backfill is incomplete |
| Numeric range filter | Min/max on every numeric column |
| Group by | `groupRows` + `useGroupedListView` ; groups closed by default, header selects the group across its pages |
| Pager | `from–to / total` , 80 rows a page, per-group pager inside an opened group |
| Row → detail | Row click opens `/resource/[id]` ; edit is a modal only ( `?add=1` , `?edit=1` ) |
| Clear all | `ClearAllModuleButton` gated on `*.bulk_delete` , typed confirmation, FK release order |
| Export | Excel/CSV reflecting the exact visible columns and order |
| Back button | Every detail route carries a Back link resolving through a per-module allowlist |

# 6 Appendices

## 6.1 Environment variables (names only)

| Group | Variables |
|---|---|
| Firebase Admin | `FIREBASE_PROJECT_ID`, `FIREBASE_CLIENT_EMAIL`, `FIREBASE_PRIVATE_KEY`, `FIREBASE_SERVICE_ACCOUNT_JSON`, `FIREBASE_API_KEY` |
| R2 | `R2_ACCOUNT_ID`, `R2_ACCESS_KEY_ID`, `R2_SECRET_ACCESS_KEY`, `R2_BUCKET_NAME`, `R2_ENDPOINT` |
| eSign worker bridge | `ESIGN_R2_WORKER_URL`, `ESIGN_R2_WORKER_SECRET` |
| Worker | `WORKER_SHARED_SECRET`, `FLEET_WS_URL`, `FLEET_WS_TOKEN_SECRET` |
| Client-exposed | `NEXT_PUBLIC_GOOGLE_MAPS_API_KEY`, `NEXT_PUBLIC_GOOGLE_MAPS_MAP_ID`, `NEXT_PUBLIC_FIREBASE_*` |
| Cron | `CRON_SECRET` |
| Feature switches | `ADMIN_API_ROUTER`, `PAGE_READ_SNAPSHOT`, `GPS_SINK`, `GPS_DUAL_WRITE`, `LOCATION_EVENTS_ARCHIVED` |
| Secrets (Secret Manager) | `SERVICE_ODOMETER_APP_SECRET` |

## 6.2 Test suites (27 `npm run test:*` scripts)

`test:geo` · `test:zones` · `test:drivers` · `test:permissions` · `test:i18n` (**12,625 keys × 2**) · `test:service` · `test:storage` · `test:assistant` · `test:recon` (57) · `test:vehicles` · `test:driver-devices` · `test:auth` · `test:performance` · `test:payroll` · `test:dpd` · `test:deliveries` · `test:earnings` · `test:visits` · `test:esign` · `test:templates` · `test:employeedesk` · `test:requests` · `test:compliance` · `test:notifications` · `test:attendance` · `test:fleet` (308) · `test:accidents` · `test:portal` · functions `npm test` (496).

## 6.3 Firestore indexes, TTL and the query budget

The database is **Enterprise edition**, which creates **no single-field index by default**. A query with no matching index is answered by reading the whole collection, billed by bytes in 4 KiB units. Every shape the code issues is therefore declared.

| Item | Count | Where |
|---|---|---|
| Composite index definitions | **429** | `firestore.indexes.json` (145 KB) |
| Distinct collection groups covered | **115** | same |
| Live indexes in the database | **431** | 429 file + 2 hand-made `drivers` indexes |
| TTL field overrides | **5** | `fieldOverrides` |
| Index definitions written by the audit script | 208 of 231 detected shapes | `npm run audit:indexes --write` |

**The fifteen collections that carry the most indexes** — the count follows the number of filter/order permutations the list screens can emit, not the table's size:

| Collection | Indexes | Why |
|---|---|---|
| `deliveries` | 127 | status × zone × partner × date × order-id prefix × shift-day, plus every sort the list offers |
| `drivers` | 17 | employee id / civil id / passport / phone prefix and the roster sorts |
| `esign_requests` | 9 | status × category × signer × due date |
| `attendance_logs` | 9 | driver × day range, zone, restaurant, status |
| `admin_activity_logs` | 9 | actor × entity × action × date |
| `notification_campaigns` | 9 | status × category × creator × schedule |
| `notification_dispatch_items` | 8 | campaign × driver × status × dismissed |
| `driver_operation_events` | 8 | driver × event × date, retention sweep |
| `driver_location_events` | 8 | driver × day, history and export windows |
| `fleet_events` | 8 | driver × type × cooldown windows |
| `driver_telemetry_events` | 8 | driver × event type × day |
| `asset_assignments` | 8 | driver × catalog × state |
| `driver_intakes` | 7 | employee id / phone / civil id lock checks |
| `order_exceptions` | 6 | rider × kind × status × day |
| `restaurants` | 6 | partner × zone × name prefix |

**TTL (all on an `expire_at` timestamp field):**

| Collection | Retention | Written by |
|---|---|---|
| `driver_telemetry_events` | 14 days | `driverIngestTelemetry` |
| `driver_operation_events` | 30 days | `log_driver_operation`, `log_driver_operation_autonomous` |
| `driver_location_events` | 90 days | `driverReportLocation`, `adminIngestDriverPositions` |
| `page_read_snapshots` | per-entry (`fresh_until`) | the scan-lane read-through cache |
| `admin_cost_samples` | sampling window | cost telemetry |

**How an index gets added.** `npm run audit:indexes` walks `functions/src` and `src` with the TypeScript compiler API, resolves every `COLLECTIONS.*` / `FIELDS.*` reference to its literal, collects each query's equality / range / order fields and checks them against an index list (the file, or a saved `firebase firestore:indexes` dump). `--write` appends the missing definitions; a single-field shape is written as `[field, __name__]` because Enterprise edition will not create one. Shapes that cannot be resolved — a field named at run time, a

query assembled filter by filter — are reported, not guessed. Deployment is `firebase deploy --only firestore:indexes --non-interactive`; indexes build online, so a query keeps scanning until its index is ready.

**Two index rules that are load-bearing, not stylistic.** A `ORDER BY created_at DESC NULLS LAST` shape needs the `NULLS LAST` twin of a plain `DESC` index — PostgREST/Firestore emit the former, and the mismatch is what once made a list degrade to a 114 MB scan. And a `driver_locations.last_seen_at` index is deliberately absent: that field is rewritten on every GPS report, so indexing it would tax the hottest write path to serve a cleanup job.

## 6.4 Cloud Function deployment inventory

| Attribute | Value |
|---|---|
| Project | `musallam-delivery-prod` |
| Region | `me-central1` (Doha) |
| Generation | Cloud Functions **gen2** (Cloud Run underneath) |
| Runtime target | Node 20 LTS ( `.nvmrc` ) |
| Callable exports | **319** across **73** module files |
| Routed exports | `adminApi` , `adminApiScan` , `adminApiHeavy` , `workerApi` |
| Scheduled exports | 5 ( `serviceOdometerSync` , `serviceVehicleDistanceSweep` , `deliveryProximitySweep` , `homeSnapshotRepair` , `bitesMirrorSweep` ) |
| HTTP exports | `workerFleetRead` , `adminIngestDriverPositions` , `adminRecordFleetEvents` (secret-gated) + `workerApi` (single door) |
| Authentication | Firebase ID token → `requireStaff` → permission slug; workers use `X-Worker-Secret` |
| Secrets | `SERVICE_ODOMETER_APP_SECRET` , `WORKER_SHARED_SECRET` (Secret Manager) |

| Lane | Memory | Concurrency | Instances | Timeout | Typical members |
|---|---|---|---|---|---|
| `adminApi` | 2 GiB | 8 | 0–8 | 60 s | Default for every admin callable not listed below |
| `adminApiScan` | 4 GiB | 2 | 0–6 | 180 s | The 19 readers in §4.1 |
| `adminApiHeavy` | 4 GiB | 1 | 0–3 | 300 s | 17 purge callables, earnings recalculation, eSign compose, import chunk claim, level sweep |
| `workerApi` | 256 MiB | 40 | warm | 120 s | The three worker doors, chosen by the last path segment |

**Deployment rules that are part of the architecture, not process preference:**

- `assertPartition()` fails the build if a callable is on two lanes or missing from an export — a name cannot be shipped unrouted.

- `laneOf(name)` is the only switch; moving a function between lanes is one edit, not a search.
- The three worker doors retain a **standby** shape (`minInstances: 0`) so a burst can still land on them after `workerApi` takes the steady traffic.
- Every lane holds a per-call `{"perf":"router"}` log line (`lane`, `fn`, `ms`, `snapshot: hit|miss`), which is how the read-snapshot hit rate is observed without adding a metric.

## 6.5 Every route under `src/app/api`

| Route | Route | Route |
|---|---|---|
| `/admin/app-releases/presign` | `/admin/driver-documents/expiry` | `/cron/roster-alerts` |
| `/admin/app-releases/register` | `/admin/driver-documents/upload` | `/cron/storage-uploads-cleanup` |
| `/admin/app-releases/setup-cors` | `/admin/locales` | `/deliveries/[id]/gps` |
| `/admin/driver-documents` | `/admin/locales/[code]` | `/deliveries/proof-download` |
| `/admin/translations/[locale]` | `/assistant/chat` | `/drivers/import-jobs` |
| `/build-id` | `/cron/attendance-auto-checkout` | `/drivers/import-template.xlsx` |
| `/cron/dpd-shift-notices` | `/cron/driver-ops-retention` | `/drivers/levels` |
| `/cron/driver-telemetry-retention` | `/cron/esign-batch-drain` | `/drivers/page` |
| `/cron/expire-stale-pickups` | `/cron/locations-cleanup` | `/driver-uploads/confirm` |
| `/cron/notifications-automations` | `/cron/notifications-dispatch` | `/driver-uploads/mine` |
| `/cron/performance-rollup` | `/cron/requests-sweep` | `/driver-uploads/presign` |
| `/driver-uploads/proxy` | `/driver-uploads/read` | `/driver-app/active-release` |
| `/driver-app/firebase-config` | `/driver-app/notification-media` | `/driver-app/notifications` |
| `/esign/batch-zip` | `/esign/document-download` | `/esign/render-bench` |
| `/launcher/weather` | `/live-tracking-v2/token` | `/menu-icons/[file]` |
| `/nav/badges` | `/notifications/events` | `/order-recon/commit` |
| `/order-recon/comparison` | `/order-recon/preview` | `/order-recon/redo` |
| `/order-recon/undo` | `/storage/signed-url` | `/usage-monitor/data` |

## 6.6 Known constraints and open items

| Item | State |
|---|---|
| Supabase Postgres | **Paused.** 398 migrations retained as history; nothing reads them |

| Item | State |
|---|---|
| BDashcam odometer feed | Endpoint configured, secret stored, function deployed — **paused** because the vendor gateway rejects `/sys/login` from our network (see `scripts/probe-bdashcam-api.mjs`) |
| Fleet Service maintenance | Rules engine shipped; **0 maintenance rules seeded in production** until `scripts/service-maintenance-seed.mjs` is applied and the baseline import re-run |
| Custom columns | Live for **Employees** and **Clients**; Restaurants, Zones and Vehicles are registered but not yet wired (`hasListColumns: false`), so the settings screen does not offer them |
| Numeric-only Excel for day cells | `combined-xlsx.ts` writes real numbers for hour cells; the CSV paths still emit strings |
| `driver_app_force_update` | Server switch live; takes effect only once a build ≥ the configured `versionCode` is on Play |
| Local `next build` | Fails in this checkout while the parallel `operations-dashboard-v2` module is untracked and incomplete; Vercel builds from the committed tree |

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*End of document.*